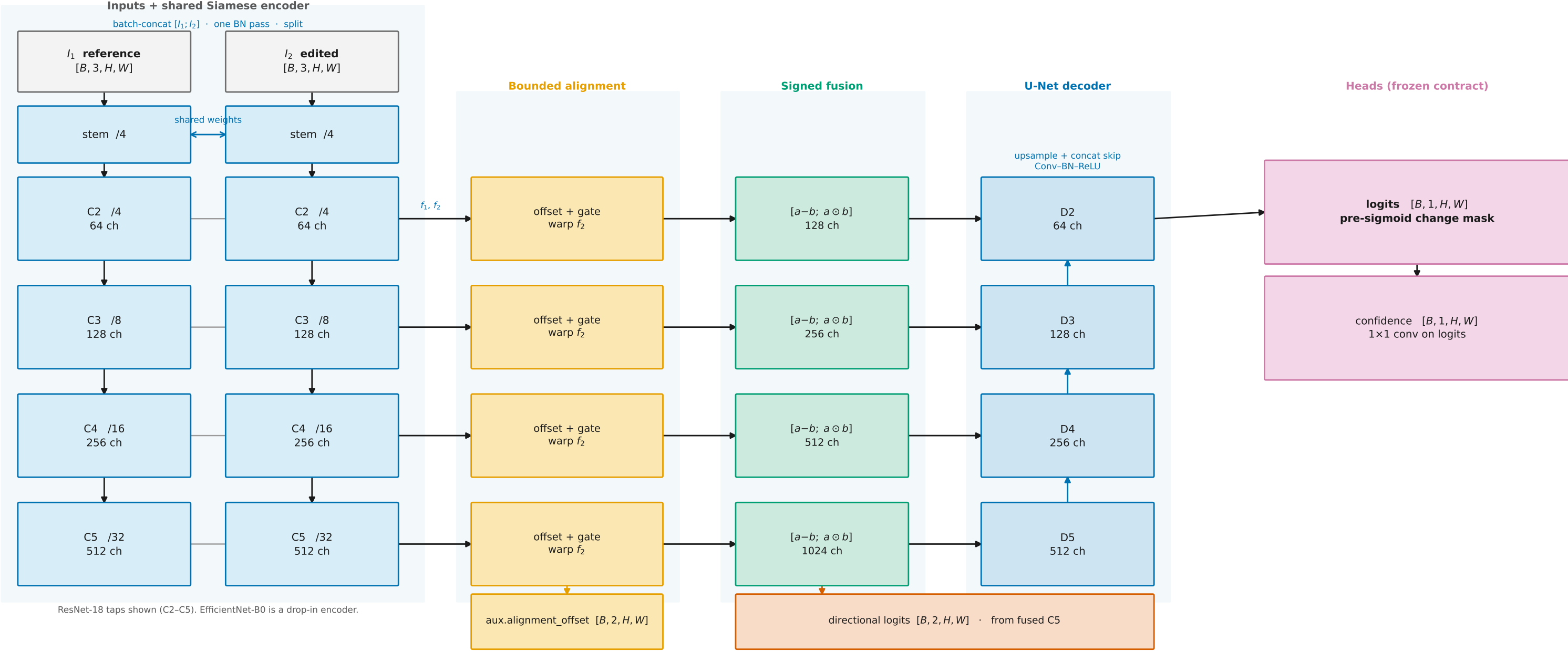
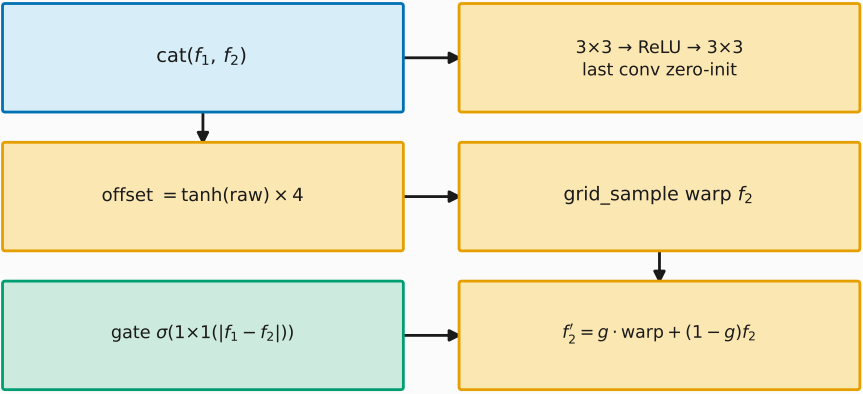


Proposed model — pair-order consistent, uncertainty-aware change detection

ProposedModel: encoder.forward_pair → bounded alignment → signed fusion → U-Net decoder → heads



(a) Bounded, change-gated alignment



Gate from abs-diff magnitude, not from the change head.

(b) Why signed fusion needs a swap loss

Default mode $\varphi(a, b) = \text{concat}(a - b, a \odot b)$

$|a - b|$ is swap-symmetric for any head h ; direction is discarded.

Signed $a - b$ is antisymmetric: $\varphi(b, a) = -\varphi(a, b)$.

$h(b - a) = h(-(a - b))$ equals $h(a - b)$ iff h is even.

A generic conv head is not even. Pair-order consistency is a learned constraint, not a freebie of weight sharing.

(c) Pair-order pass at train time

